



LLM-based Academic Writing with Fine-Grained Citations

Master Thesis Presentation

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October 16, 2025, Niklas Munkes

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dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple neural architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrent and convolutional layers. Experiments on machine translation show superior to quality while being more parallelizable and faster to train. Our model achieves 28.4 BLEU on German translation task, improving over the previous best by over 2 BLEU. Our 59.5M parameter model establishes a new single-model state-of-the-art on eight NLP tasks, including training for 3.5 days on eight GPUs, a small fraction of the compute of the previous best. We show that the model works by applying it successfully to English

While the Transformer architecture has become the de-facto standard for natural language processing tasks, its applications to computer vision remain limited. In this work, attention is either applied in conjunction with convolutional neural networks or used to replace certain components of convolutional neural networks. We show that this red and a pure transformer applied directly to sequences very well on image classification tasks. When data are transferred to multiple end-to-end or small (ImageNet, CIFAR-100, etc.) datasets, the results compared to state-of-the-art convolutions are significantly lower computational resources in training.

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In this report, we introduce *Quest25*, a comprehensive suite of large language models (LLMs) designed to meet diverse needs. Compared to previous iterations, *Quest25* has been significantly improved during both the pre-training and post-training stages. In terms of pre-training, we have scaled the high-quality pre-training datasets from the previous 100 billion tokens to 1.5 trillion tokens, and introduced a new data source, Common Sense expert knowledge, and training capabilities. In terms of post-training, we implement iterative supervised fine-tuning with over 1 trillion tokens, as well as multi-stage reinforcement learning, including offline learning DPO and online learning CKPO. Fine-tuning techniques significantly enhance human performance, and results have not only surpassed standard data sets, but also broken the existing SOTA.

To handle discrete and variational cases effectively, we present DynaV2 LHM series in both configurations. The experimental results on both base models and instruction-tuned models in sizes of 0.5B, 1.8B, 7B, 14B, 33B, and 72B parameters. Quantized versions of the instruction-tuned models are also provided. Over 100 models can be accessed from Hugging Face Hub, ModelScope, and Kaggle. In addition, for limited situations, the proprietary models currently include fine-tuning of experts (MultiLoRA) and CoT (Chain-of-Thought) and CoV2 (Chain-of-Values) are available from <https://huggingface.co/01-ai>.

Ques2.5 has demonstrated top-first performance in a variety of large-scale understanding, reasoning, and commonsense tasks, significantly outperforming the open-weight flagship GPT-4o models. Ques2.5 is a member of the Qwen2.5 family, which includes a number of other open and proprietary models. The main focus of the article is the top-first model, Qwen2.5-72B, but larger Qwen2.5-72B and Qwen2.5-72B-Chat are also available, competing against GPT-4o mini and GPT-4o in the same format. Qwen2.5 models have been implemented in the Qwen2.5-Chat (Yang et al., 2024a), Qwen2.5-Chat (Jiang, 2024a), and multimodal models.

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The dominant sequence-to-sequence models are based on complex architectures such as convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple neural network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. On all tasks we evaluated, the Transformer performed as well as or better than the best sequence-to-sequence models, despite being superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-French translation task, improving over the existing best models, including multi-headed attention models. On the WMT 2016 English-to-German translation task, our model achieves a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with

Recurrent neural networks, long short-term memory [13] and gated recurrent [14] neural networks in particular, have been firmly established as state-of-the-art approaches in sequence modeling and

³Significance comparison. Learning results in random, low-frequency, repeating RNNs with self-attention and vector embeddings for the previous time steps. As noted with Liko, developing and implementing this Transformer model and architecture has been crucially involved in every aspect of this work. No prior research could do product attention, multi-head attention and the parameter-free position representation and because the other papers involved in nearly every detail. Also, deep learning, implemented, based on evaluated continuous model variants in our original codebase and transformer. Liko also experimented with novel model variants, was responsible for our initial codebase, and efficient inference and visualizations. Likewise and Andrew spent countless hours doing time-consuming parts of our implementing transformer, replacing our earlier codebase, greatly improving results and modifying architecture.

[†]Work performed while at Google Brain.

23rd Conference on Neural Information Processing Systems (NIPS 2017), Long Beach, CA, USA.

Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that estimate the output. The first Transformer model is a simple encoder-decoder architecture. The Transformer model consists of a stack of self-attention layers. The Transformer model is trained on a large corpus of text. The Transformer model achieves state-of-the-art results on a variety of tasks, including machine translation, text summarization, and question answering.

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Attention is All You Need

Published as a conference paper at ICLR 2017

An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale

OpenAI

Qwen2.5 Technical Report

OpenAI

Abstract

In this report, we introduce Qwen2.5, a comprehensive suite of large language models (LLMs) designed to meet diverse needs. Compared to previous models, Qwen2.5 has significantly improved in both the pre-training and post-training stages. In terms of pre-training, we have scaled the high-quality pre-training dataset from the previous 7 trillion tokens to 18 trillion tokens. This provides a stronger foundation for various tasks, open knowledge, and reasoning capabilities. In terms of post-training, we implement various post-training techniques, such as iterative sampling, as well as various reinforcement learning techniques, to further improve the model's performance. Qwen2.5 has achieved state-of-the-art results on a wide range of tasks, including machine translation, text summarization, and question answering.

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1. INTRODUCTION

3.1. Vision Transformer (ViT)

Self-supervised pre-training

Overview of the model is depicted in Figure 2. The standard Transformer receives an input X of size $N \times D$, where N is the number of tokens and D is the dimensionality of the embedding.

3. Pre-training

Our language model pre-training process consists of several key components. First, we carefully curate a large-scale dataset of text and code. This dataset is then processed into tokens and embeddings. The pre-training process involves training the model on this dataset using a self-supervised objective. The model is trained to predict missing tokens in the input sequence. The pre-training process is divided into two main stages: pre-training and post-training. The pre-training stage involves training the model on the large-scale dataset. The post-training stage involves fine-tuning the model on specific tasks.

3.1. Vision Transformer (ViT)

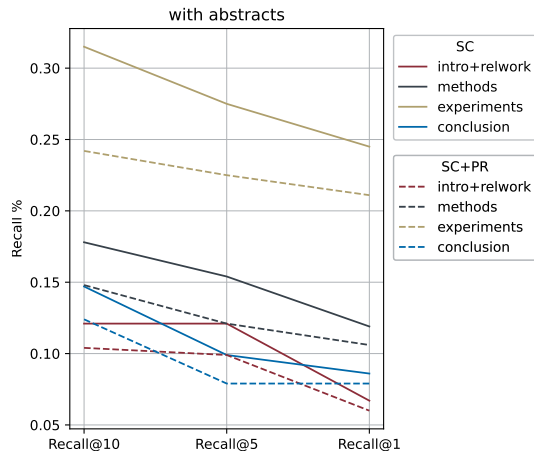
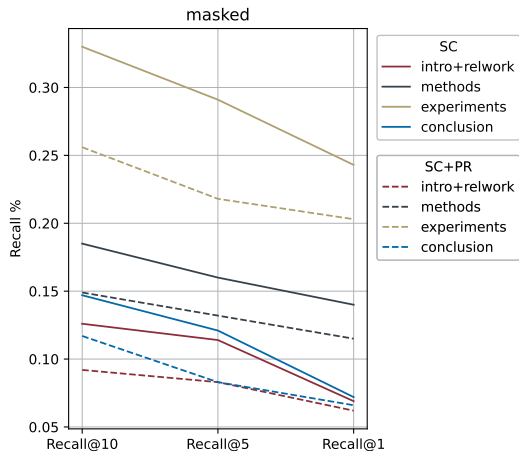
The Vision Transformer (ViT) is a deep learning architecture for image classification. It is based on the Transformer architecture. The ViT takes an input image and divides it into patches. These patches are then embedded into a sequence of tokens. The ViT uses a self-supervised pre-training process to learn the relationships between the patches. The ViT is trained to predict missing patches in the input sequence. The ViT achieves state-of-the-art results on a variety of image classification tasks.

Figure 1: The development of the Qwen series. Data collected through a search of OpenAI GPT-3 models released between 2020 and 2023. The chart shows the number of tokens used for pre-training and the number of parameters. The chart shows that the number of tokens used for pre-training has increased significantly over time, and the number of parameters has also increased. The chart shows that the Qwen series models have achieved state-of-the-art results on a variety of tasks.

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Citation Recall Performance





Recall@ k decline towards small k

Model	Dataset	intro+relwork	methods	experiments	conclusion	Σ
SC	masked	-0.06	-0.04	-0.09	-0.07	-0.26
SC	with-abstracts	-0.05	-0.06	-0.07	-0.06	-0.24
SC+PR	masked	-0.03	-0.03	-0.05	-0.05	-0.16
SC+PR	with-abstracts	-0.04	-0.04	-0.03	-0.04	-0.15

Table: Cumulative Recall@ k difference from $k = 10$ over $k = 5$ to $k = 1$.



Why is the recall for experiments so much higher?

	intro+relwork	methods	experiments	conclusion
last token in abstract	0.08	0.12	0.22	0.11

Table: Fraction of cases where the last token before the citation appears in the abstract of the cited document.

Correct Retrieval Overlap

Dataset	Metric	intro+relwork	methods	experiments	conclusion	μ
masked	Recall@10	0.91	0.97	0.98	0.91	0.943
masked	Recall@5	0.94	0.94	0.96	0.94	0.945
masked	Recall@1	0.79	0.89	0.88	0.88	0.86
with-abstracts	Recall@10	0.9	0.93	0.96	0.95	0.935
with-abstracts	Recall@5	0.86	0.97	0.95	0.89	0.918
with-abstracts	Recall@1	0.87	0.87	0.93	0.84	0.878

Table: Fraction of correct retrievals by ScholarCopilot given correct retrieval by the PR module.



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